Computer Vision – 21CSE454T

Unit 3 - Session 8

SLO 2: GHT for Feature Collation & Advanced Topics- Critique GHT's scalability limitations

Duration: 10 Minutes Activity

Question:

Study all 6 cards before filling in the ranking table.

MEM B1

Memory Explosion *[Curse of Dimensionality]*

The accumulator size grows multiplicatively with each new parameter dimension. A 5D ellipse accumulator (cx, cy, a, b, theta) can require hundreds of millions of bins, consuming gigabytes of RAM.

Impact: *Each added parameter multiplies total memory by the number of bins in that dimension.*

TIME B2

Voting Time *[Computational Complexity]*

Every boundary pixel must cast votes across all valid parameter combinations. With thousands of edge pixels and millions of accumulator bins, the total voting operations per frame can reach billions.

Impact: *Voting complexity = O(N x M) where N = edge pixels, M = valid parameter combinations per pixel.*

BIN B3

Bin Resolution Trade-off *[Precision vs Speed]*

Fine bins give precise estimates but increase memory and cost. Coarse bins are faster but may merge nearby objects into one peak or miss fine positional details.

Impact: *Halving bin size in 5D increases total bins by 2^5 = 32 times.*

SCL B4

Multi-Scale Search *[Scale Invariance Cost]*

If object size is unknown, the system must search across a range of scales. Each scale level creates a separate accumulator, multiplying total computation.

Impact: *S scale levels multiply total votes and memory by factor S.*

CLT B5

Cluttered Scenes *[False Peak Density]*

When many other edges exist in the image (eyelashes, glare, reflections, glasses), spurious boundary points cast votes that create false peaks, reducing detection reliability.

Impact: *False peaks compete with the true peak; selection becomes unreliable in dense clutter.*

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ROT B6

Rotation Invariance Cost *[Orientation Search Space]*

To detect objects at any orientation, the system must maintain separate R-tables per orientation bin or expand the accumulator with a theta dimension, increasing memory and voting cost.

Impact: *R orientation bins add one full dimension, multiplying accumulator size by R.*

Rank all 6 bottlenecks for a real-time iris detection system. Write the ID (B1-B6) and name in column 2, and give a one-sentence reason in column 3.

Rank	Bottleneck ID & Name	Brief reason — one sentence
1 Most Critical	B1 — Memory Explosion	A 5D accumulator can require hundreds of millions of bins, causing extreme memory use and making real-time processing difficult.
2	B2 — Voting Time	Voting across thousands of edge pixels and millions of parameter combinations can require billions of operations per frame.
3	B4 — Multi-Scale Search	Searching unknown object sizes creates separate accumulators for each scale, multiplying both computation and memory.
4	B6 — Rotation Invariance Cost	Adding orientation bins creates another accumulator dimension, multiplying memory and voting cost.
5	B3 — Bin Resolution Trade-off	Fine bins improve precision but greatly increase memory and computation, while coarse bins can reduce detection precision.
6 Least Critical	B5 — Cluttered Scenes	Unrelated edges can create false peaks and reduce detection reliability, especially in cluttered iris images.